Problem 1: Trotterized Quantum Simulation

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Abstract

Trotterization is one of the most widely used approaches to implementing Hamiltonian time evolution on a quantum computer. By decomposing the exponential of a Hamiltonian into a product of simpler exponentials, this practical method for quantum simulation exhibits a trade-off between approximation accuracy and circuit depth. Although this asymptotic trade-off is suboptimal compared with quantum singular value transformation, its simplicity at small scales makes it suitable for applications in the early fault-tolerant regime. In this challenge, participants will study Trotterized time evolution for a periodic Heisenberg model, analyze the resulting circuit structure, and investigate how spectral information can be extracted from the dynamics. Teams are encouraged to build on the official quantum simulation materials, replacing the tutorial Hamiltonian with the Heisenberg model below and exploring it creatively. The project may include simulation, circuit optimization, controlled time evolution, or spectral estimation based on autocorrelation signals.

$$-i\frac{d}{dt}|\psi\rangle = H|\psi\rangle \quad \Rightarrow \quad |\psi(t)\rangle = e^{-iHt}|\psi_0\rangle \quad \Rightarrow \quad \boxed{U(t) = e^{-iHt}}$$

$$|\psi(t)\rangle = U(t)|\psi_0\rangle$$

Exact Simulation is harder to calculate as N(qubit number grows)
($\dim(H) = 2^N$**)**

➔ Solution: Trotterization!

Hamiltonian over entire time is divided into tiny steps of state 'evolution'.

$$e^{-iHt} \approx \left(\Pi_l e^{-\frac{iH_l t}{k}}\right)^k = \left[\left(e^{-\frac{iH_1 t}{k}}\right)\left(e^{-\frac{iH_2 t}{k}}\right)\dots\left(e^{-\frac{H_l t}{k}}\right)\right]^k$$

k:number of repetitions of cycle
l: number of Hamiltonian in a cycle

where $\frac{t}{k}$ **is considered 'one' step &** $H = \Sigma_l H_l$

However… $e^{-iH_1 t}e^{-iH_2 t} \neq e^{-(H_1+H_2)t}$ **because** $H_1 \& H_2$ **are noncommutative!**

$$[H_1, H_2] = H_1 H_2 - H_2 H_1 \neq 0$$

However… if $A \& B$ **are non commutative, by BCH formula,**

$e^A e^B = e^C$ **where** $C = A + B + \frac{1}{2}[A,B] + \frac{1}{12}\big[A,[A,B]\big] + \frac{1}{12}\big[B,[B,A]\big] + \cdots$

$e^A e^B e^C = e^D$ **where** $D = A + B + C + \frac{1}{2}([A,B] + [A,C] + [B,C]) + \cdots$

So Trotterization has certain errors…

The Hamiltonian Model we use is

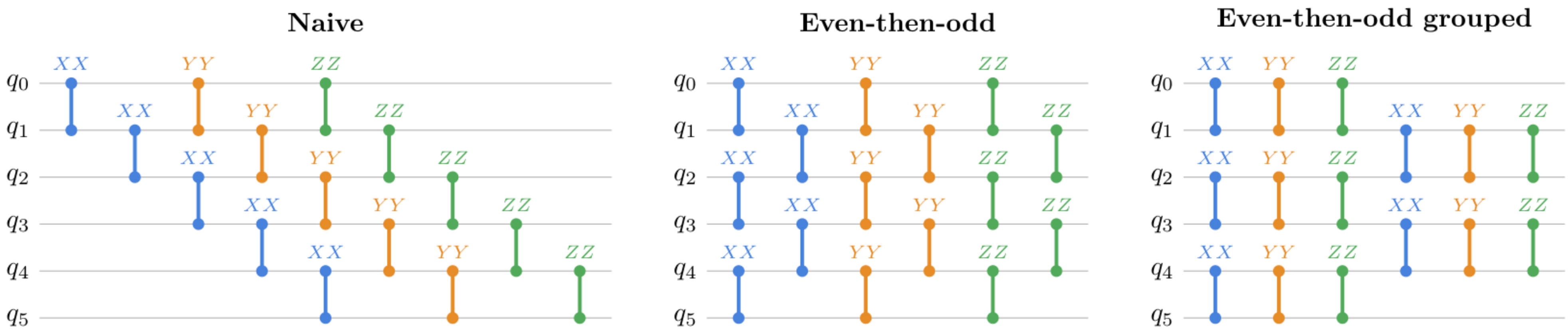
$H = \Sigma_{j=0}^{N-1}(J_x X_j X_{j+1} + J_y Y_j Y_{j+1} + J_z Z_j Z_{j+1}) + h_x \Sigma_{j=0}^{N-1} X_j$ where $j+1$ is modulo N

$N = 6, J_x = 1.0, J_y = 0.7, J_z = 1.2, h_x = 0.4$ & $|\psi_0\rangle = |010101\rangle$

X, Y, Z are called Pauli Terms.

➔ The Hamiltonian is periodic & anistropic.

Following Orderings are possible



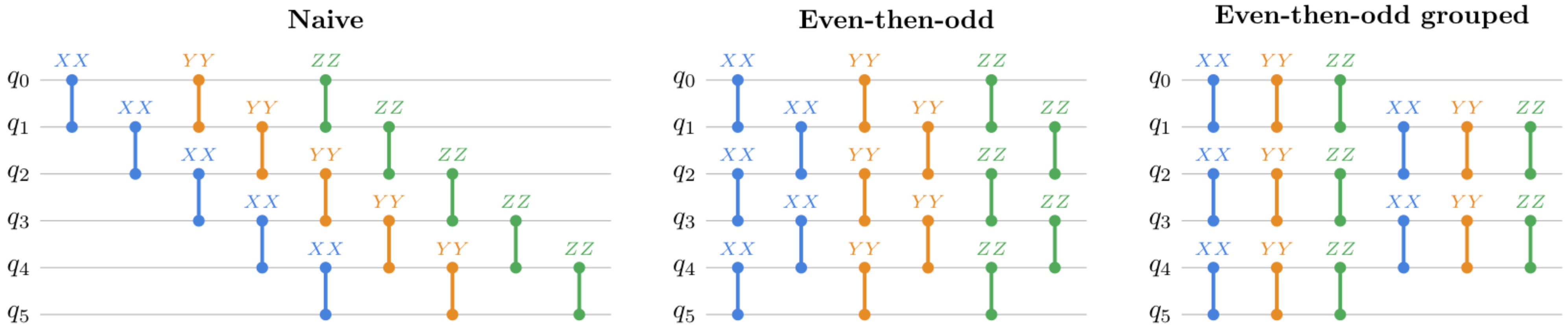
(a) Construct Hamiltonian using 'SparsePauliOp'. Report total number of Pauli terms and identify which terms come from the periodic boundary condition.

Hint: Refer to the following figure to optimize the circuit depth and expected Trotter error. Explain which specific term ordering is expected to give lower circuit depth and Trotter error, and why.

$H = \Sigma_{j=0}^{N-1}\left(J_x X_j X_{j+1} + J_y Y_j Y_{j+1} + J_z Z_j Z_{j+1}\right) + h_x \Sigma_{j=0}^{N-1} X_j$ **where $j+1$ is modulo N**

Pauli Terms: 6+6+6+6=24

Periodic Boundary: $X_0 X_5, Y_0 Y_5, Z_0 Z_5$



Error = Sum of non-commuting combinations

(a) Construct Hamiltonian using 'SparsePauliOp'. Report total number of Pauli terms and identify which terms come from the periodic boundary condition.

Hint: Refer to the following figure to optimize the circuit depth and expected Trotter error. Explain which specific term ordering is expected to give lower circuit depth and Trotter error, and why.

Lower circuit depth: Even then odd (w&w/o grouping)

➔ Reason: Even-then-odd allows for simultaneous computation in circuit!

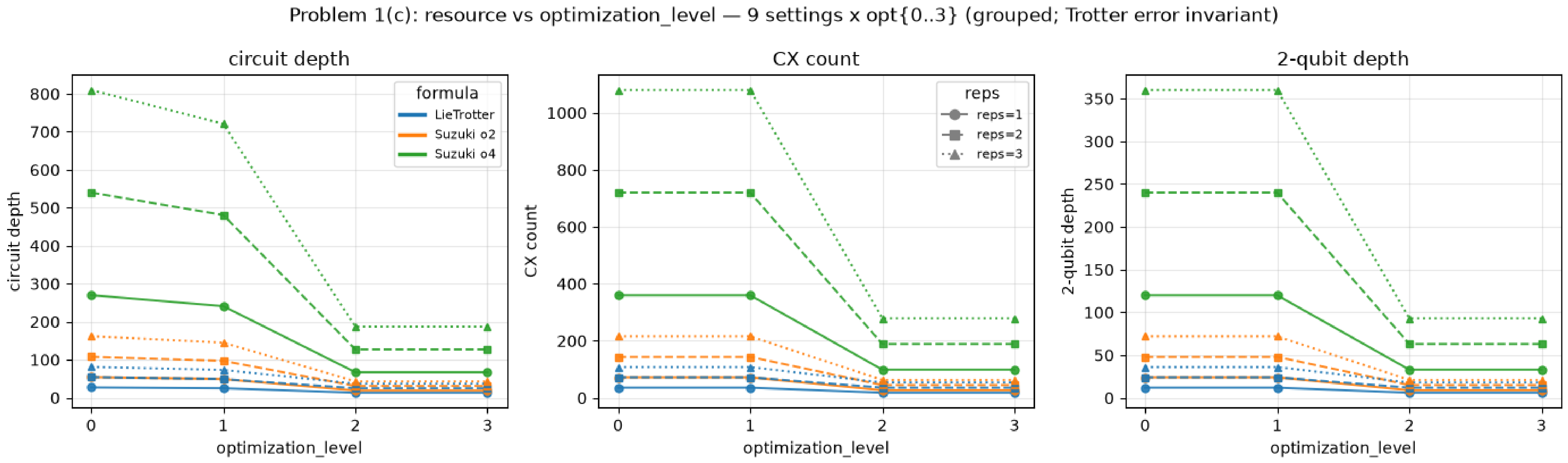
ordering	opt0_depth	opt0_cx	opt2_depth	opt2_cx
naive	79	36	37	18
even_then_odd	27	36	22	36
grouped	27	36	13	18

Even-then-odd grouping is lowest trotter error

➔ Reason: Grouping allows for more optimality as optimization_level rises (>=2) due to KAK gate synthesis

Optimization level 0, 1, 2, 3

KAK ← CX/gate (opt≥2)
same bond commuting R_{XX}, R_{YY} , R_{ZZ} -> one 2q block



Various Trotterization Formulas Exist

Lie Trotter (first-order): $S_1\left(\frac{t}{k}\right) = \Pi_{l=1}^{L} e^{-\frac{iH_l t}{k}}$ **&** $U(t) \approx \left[S_1\left(\frac{t}{k}\right)\right]^k$

2nd order Suzuki-Trotter: $S_2\left(\frac{t}{k}\right) = \left(\Pi_{l=1}^{L} e^{-\frac{iH_l t}{2k}}\right)\left(\Pi_{l=L}^{1} e^{-\frac{iH_l t}{2k}}\right)$ **&** $U(t) \approx \left[S_2\left(\frac{t}{k}\right)\right]^k$

Suzuki Formulas are Recursive

If $p = \frac{1}{4-4^{\frac{1}{3}}}$

4th order Suzuki-Trotter: $S_4\left(\frac{t}{k}\right) = S_2\left(p\,\frac{t}{k}\right)^2 S_2\left(\frac{(1-4p)t}{k}\right) S_2\left(p\,\frac{t}{k}\right)^2$ **&** $U(t) \approx \left[S_4\left(\frac{t}{k}\right)\right]^k$

General Formula: for even order $2m$**,**

$$S_{2m}\left(\frac{t}{k}\right) = S_{2m-2}\left(p_m\,\frac{t}{k}\right)^2 S_{2m-2}\left((1-4p_m)\,\frac{t}{k}\right) S_{2m-2}\left(p_m\,\frac{t}{k}\right)^2 \text{ \& } U(t) \approx \left[S_{2m}\left(\frac{t}{k}\right)\right]^k$$

where $p = \frac{1}{4-4^{\frac{1}{2m-1}}}$

General Formula: for even order $2m$,

$$S_{2m}\left(\frac{t}{k}\right) = S_{2m-2}\left(p_m \frac{t}{k}\right)^2 S_{2m-2}\left((1-4p_m)\frac{t}{k}\right) S_{2m-2}\left(p_m \frac{t}{k}\right)^2 \text{ \& } U(t) \approx \left[S_{2m}\left(\frac{t}{k}\right)\right]^k$$

where $p = \frac{1}{4-4^{\frac{1}{2m-1}}}$

Error is discrepancy:

$$\epsilon = \left| U(t) - \left[S_{2m}\left(\frac{t}{k}\right)\right]^k \right| \approx k \times O\left(\frac{t^{2m+1}}{k^{2m+1}}\right) = O\left(\frac{t^{2m+1}}{k^{2m}}\right)$$

(b) Construct product-formula circuits for each target time t in the time grid. For each t, directly approximate $U(t) = e^{-iHt}$ using 'PauliEvolutionGate' with given synthesis options for $k = 1, 2, 3$

(c) Decompose or transpile each circuit into the basis gates $\{cx, u3, u1\}$. For each synthesis option, report circuit depth, total gate count, number of cx gates, and number of one-qubit gates.

- Opt-level=3
- t=0.5
- Grouped Ordering
- N=6

Synthesis Options:

(as given by problem)

```
LieTrotter(reps=k),
SuzukiTrotter(order=2, reps=k),
SuzukiTrotter(order=4, reps=k).
```

Product-Formula Circuit Resource Counts

setting	depth	gate count	two-qubit gates	one-qubit gates
LT_k1	13	56	18	38
ST2_k1	19	87	27	60
ST4_k1	67	300	99	201
LT_k2	25	113	36	77
ST2_k2	31	138	45	93
ST4_k2	127	570	189	381
LT_k3	37	168	54	114
ST2_k3	43	192	63	129
ST4_k3	187	837	279	558

02 Problem 1

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```

Trotter Order

reps	formula	depth	gate count	2q gates	1q gates	avg ratio (vs LT)
1	LT	27	108	36	72	1
1	ST2	54	215	72	143	1.99
1	ST4	270	1075	360	715	9.97

$$S_2 \approx 2S_1$$

$$S_4 \approx 5S_2 \approx 10S_1$$

Repetition Number

formula	k=1	k=2	k=3	trend
LT	36	72	108	linear
ST2	72	144	216	linear
ST4	360	720	1080	linear

Resource count $\propto k$

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```
LieTrotter(reps=k),
SuzukiTrotter(order=2, reps=k),
SuzukiTrotter(order=4, reps=k).
```

Transpile-induced Resource Reduction

option	2q gates opt0	2q gates opt3	reduction
LT_k1	36	18	50.00%
ST2_k1	72	27	62.50%
ST4_k1	360	99	72.50%
LT_k3	108	54	50.00%
ST2_k3	216	63	70.80%
ST4_k3	1080	279	74.20%

- optimization greatly reduces 2q gates
- reduction stronger for higher-order Suzuki
- ST4 remains resource-heavy even after optimization

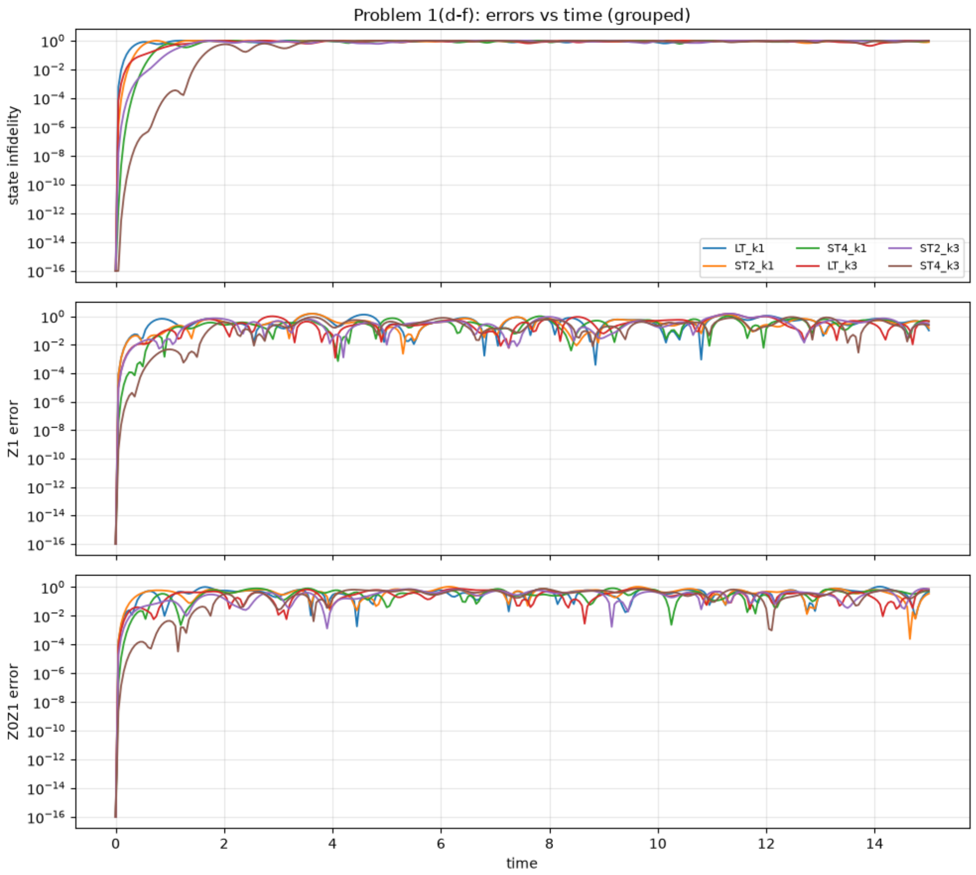
02 Problem 1

- Opt-level=3
- t=0.5
- Grouped Ordering
- N=6

(d) Use exact classical simulation for $N = 6$ to obtain the reference state $|\psi_{exact}(t)\rangle = e^{-iHt}|\psi_0\rangle$. Compare it with the product-formula state $|\psi_{PF}(t)\rangle$ using the state infidelity $\epsilon_{state}(t) = 1 - |\langle\psi_{exact}(t)|\psi_{PF}(t)\rangle|^2$.

(e) Compute local-observable errors for Z_1, Z_0Z_1. For an observable O, use $\epsilon_O(t) = |\langle O\rangle_{exact}(t) - \langle O\rangle_{PF}(t)|$.

(f) Plot time versus error for the state infidelity, Z_1 error, and Z_0Z_1 error. Compare Lie-Trotter, 2nd order Suzuki-Trotter, and 4th order Suzuki-Trotter formulas.



Short-time product-formula scaling

$$\epsilon_{\text{state}} \sim \frac{t^{2(p+1)}}{r^{2p}}$$

- Higher Trotter order ← delayed error accumulation
- Larger reps ← stronger error suppression
- Short-time product-formula scaling
- t^*: first crossing of $\epsilon_{state} > 10^{-2}$

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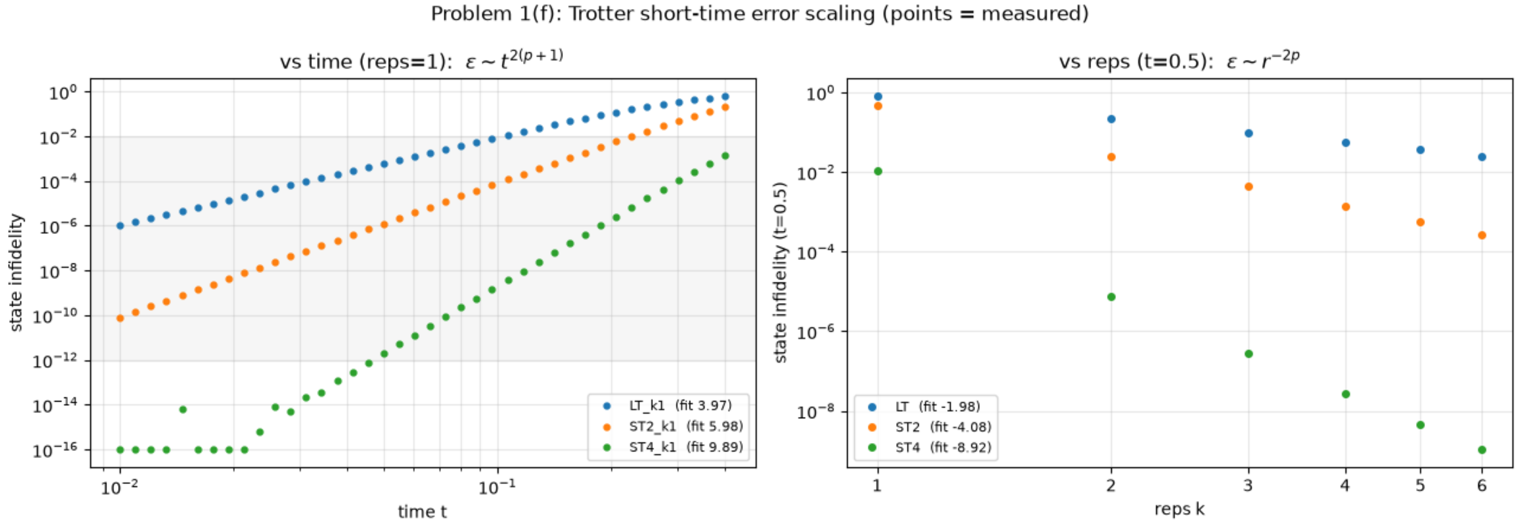

formula	time slope	theory	time agreement	reps slope	theory	reps agreement
LT	3.97	4	99.30%	-1.98	-2	99.00%
ST2	5.98	6	99.70%	-4.08	-4	98.00%
ST4	9.89	10	98.90%	-8.92	-8	88.50%

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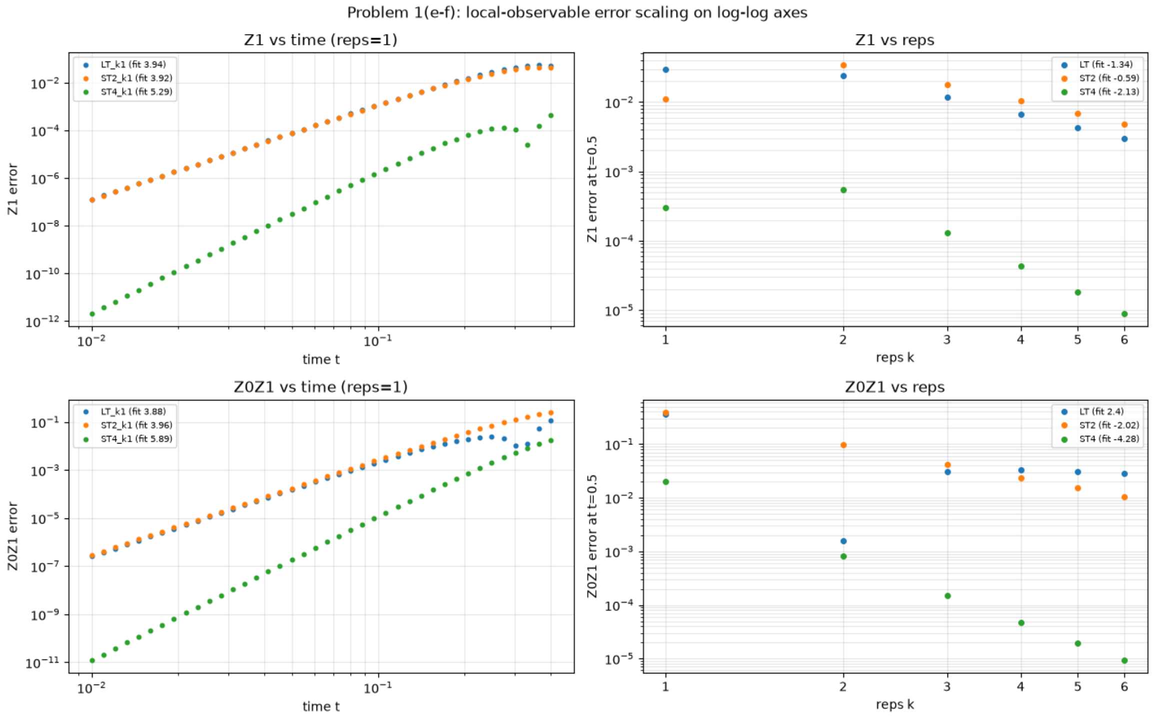

observable	setting	time slope	time theory	reps slope	reps theory
Z1	LT_k1	3.94	2	-1.34	-1
Z1	ST2_k1	3.92	3	-0.59	-2
Z1	ST4_k1	5.29	5	-2.13	-4
Z0Z1	LT_k1	3.88	2	2.4	-1
Z0Z1	ST2_k1	3.96	3	-2.02	-2
Z0Z1	ST4_k1	5.89	5	-4.28	-4

Observable slopes are more sensitive to cancellation and fitting-window choice

- Opt-level=3
- t=0.5
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- N=6

(e) Compute local-observable errors for Z_1, Z_0Z_1. For an observable O, use $\epsilon_O(t) = |\langle O\rangle_{exact}(t) - \langle O\rangle_{PF}(t)|$.

(f) Plot time versus error for the state infidelity, Z_1 error, and Z_0Z_1 error. Compare Lie-Trotter, 2nd order Suzuki-Trotter, and 4th order Suzuki-Trotter formulas.

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad Z_1 = I \otimes Z \otimes I \otimes I \otimes I \otimes I$$

$$\langle Z_1\rangle_{exact}(t) = \langle\psi(t)|Z_1|\psi(t)\rangle = \langle\psi_0|e^{iHt}Z_1e^{-iHt}|\psi_0\rangle$$

$$\langle Z_0Z_1\rangle_{exact}(t) = \langle\psi(t)|Z_0Z_1|\psi(t)\rangle = \langle\psi_0|e^{iHt}Z_0Z_1e^{-iHt}|\psi_0\rangle$$

If we let $|\psi(t)\rangle = \Sigma_{b\in\{0,1\}^6} c_b(t)|b_0b_1b_2b_3b_4b_5\rangle = \Sigma c_b(t)|b\rangle$,

$$c_b(t) = \langle b|e^{-iHt}|010101\rangle \qquad \text{Since } |\psi_0\rangle = 010101\rangle$$

$$Z_1|b_0b_1b_2b_3b_4b_5\rangle = (-1)^{b_1}|b_0b_1b_2b_3b_4b_5\rangle$$

Thus $\langle Z_1\rangle_{exact}(t) = \Sigma_{b\in\{0,1\}^6}(-1)^{b_1}|c_b(t)|^2 = P(q_1 = 0) - P(q_1 = 1)$

- Opt-level=3
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Thus $\langle Z_1 \rangle_{exact}(t) = \Sigma_{b\in\{0,1\}^6}(-1)^{b_1}|c_b(t)|^2 = P(q_1 = 0) - P(q_1 = 1)$

In the same way,

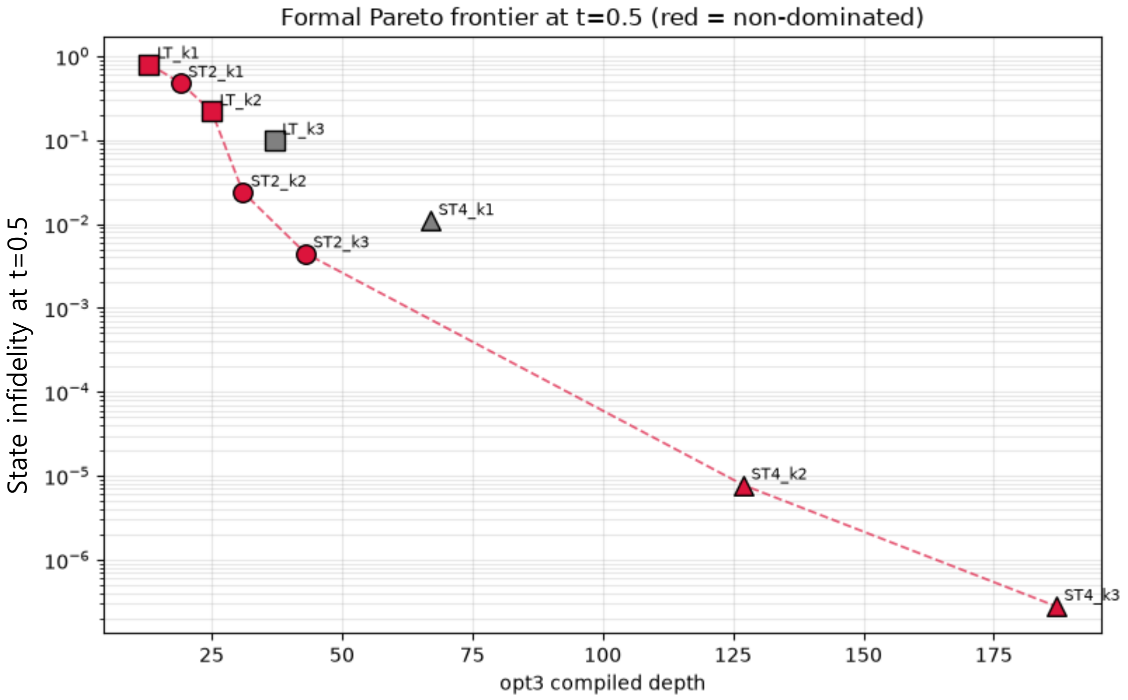
$$Z_0Z_1|b_0b_1b_2b_3b_4b_5\rangle = (-1)^{b_0+b_1}|b_0b_1b_2b_3b_4b_5\rangle$$

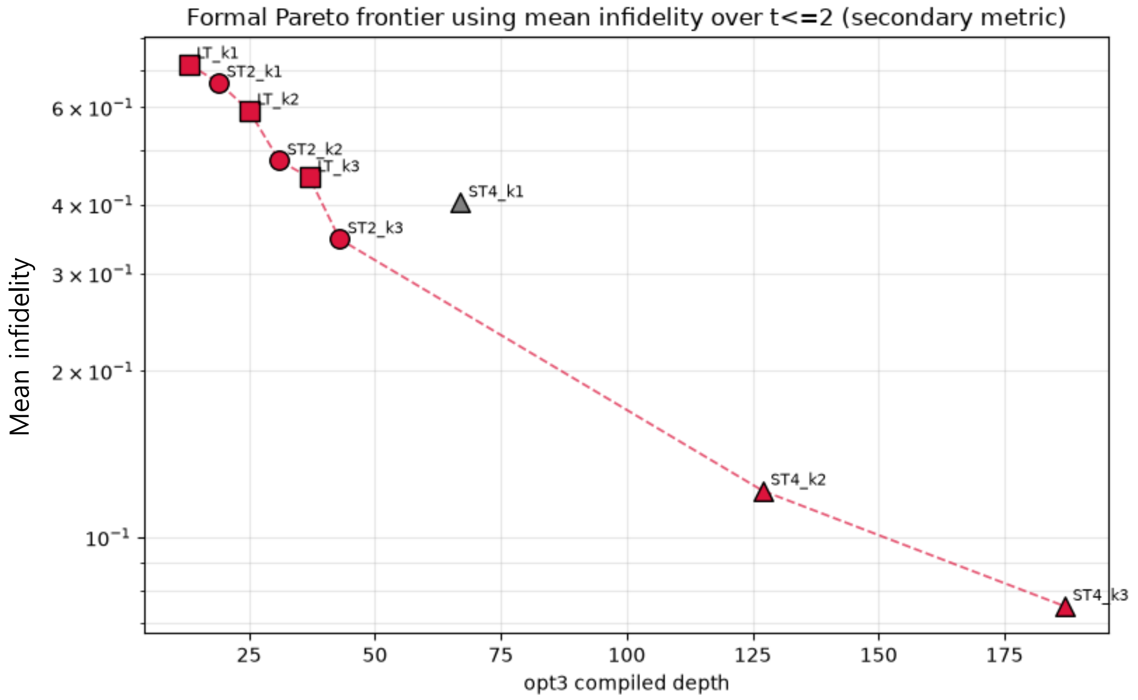
$$\langle Z_0Z_1 \rangle_{exact}(t) = \Sigma_{b\in\{0,1\}^6}(-1)^{b_0+b_1}|c_b(t)|^2 = P(q_1 = q_0) - P(q_1 \neq q_0)$$

$\langle Z_1 \rangle$ measures imbalance between qubit 1 being 0 or 1

$\langle Z_0Z_1 \rangle$ measures whether qubits 0 and 1 are aligned or opposite

(g) Discuss the trade-off between approximation accuracy and circuit depth. Which construction is best for the same repetition number? Which construction is best for a similar circuit depth?





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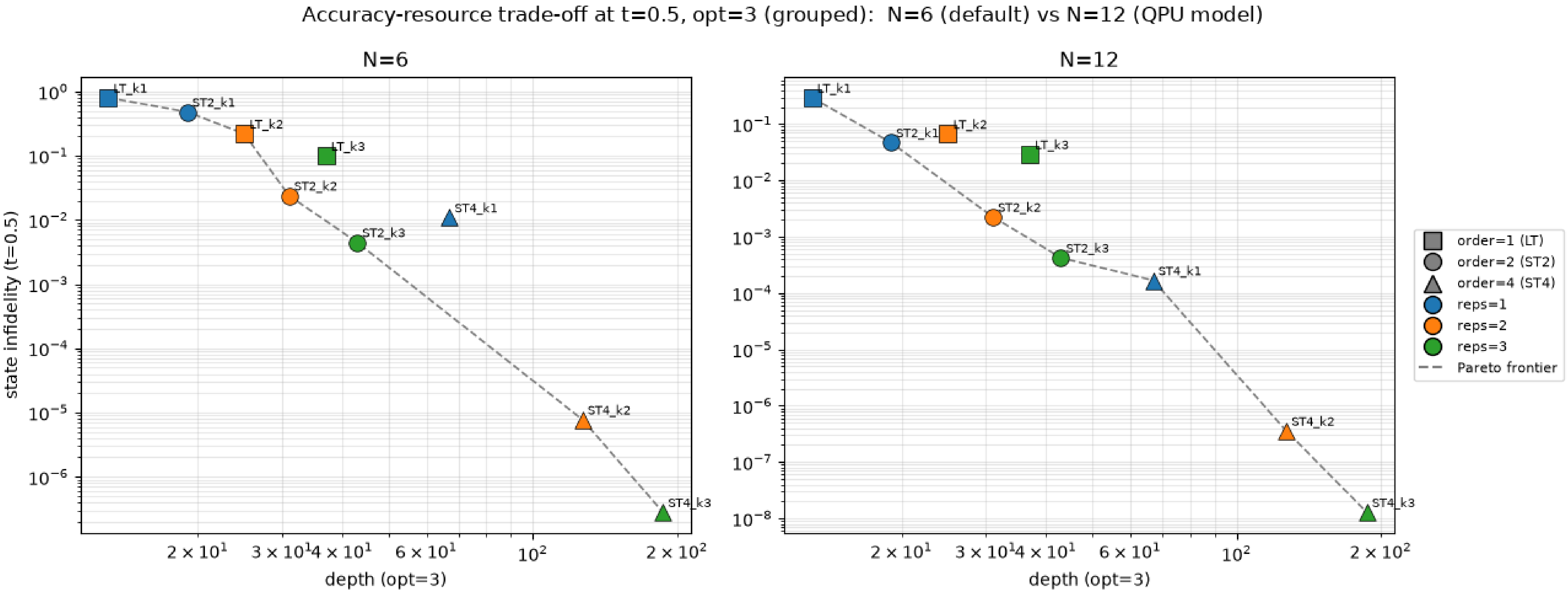

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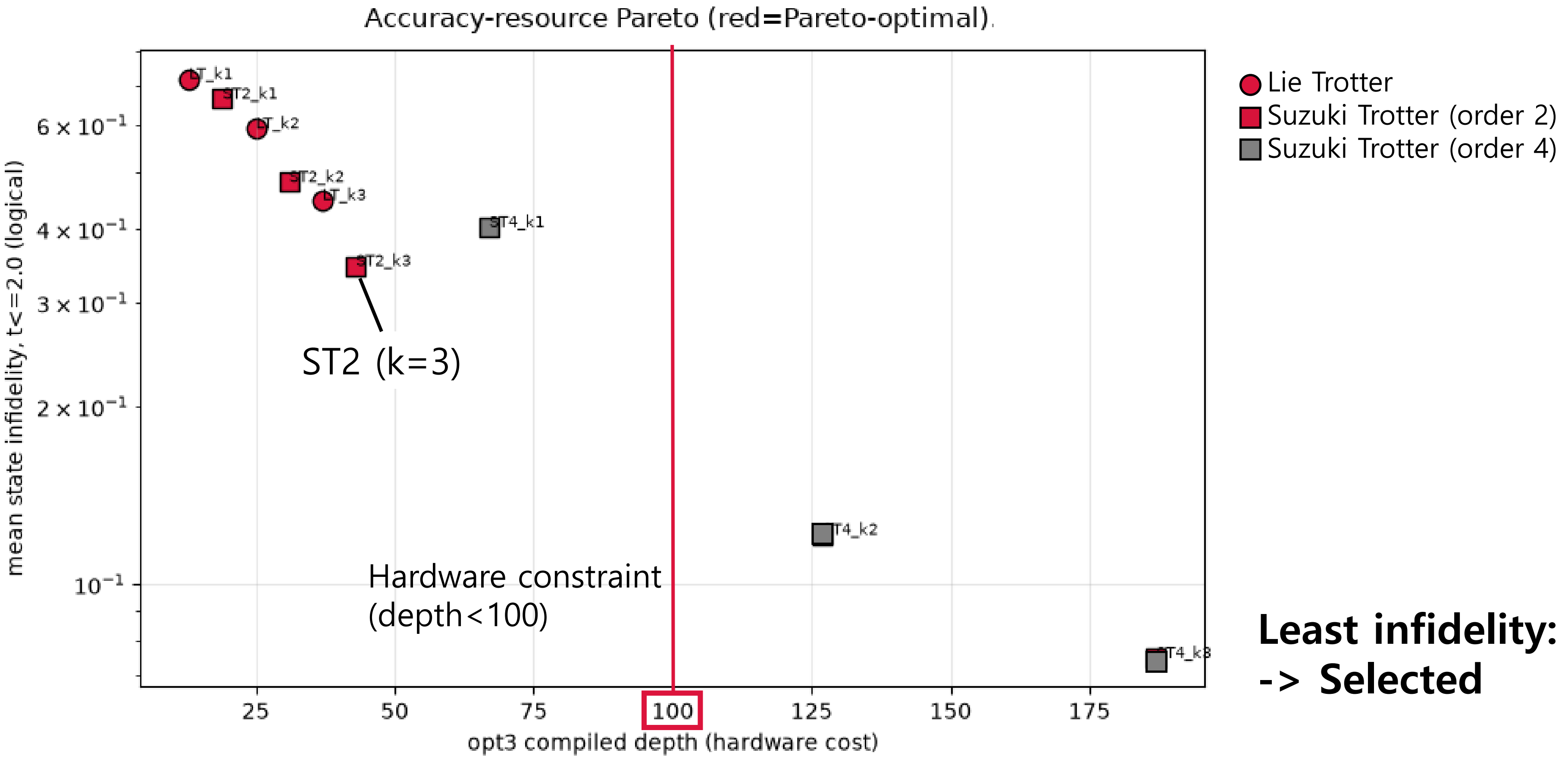


Least infidelity: Suzuki Trotter (order 2)
-> Selected

(g) Discuss the trade-off between approximation accuracy and circuit depth. Which construction is best for the same repetition number? Which construction is best for a similar circuit depth?

[CX Gate Count] ordering='even_odd_grouped', opt_level=3

model	N=12	N=20	N=28	N=36	N=44	N=52
LT_k1	36	60	84	108	132	156
LT_k2	72	120	168	216	264	312
LT_k3	108	180	252	324	396	468
ST2_k1	54	90	126	162	198	234
ST2_k2	90	150	210	270	330	390
ST2_k3	126	210	294	378	462	546
ST4_k1	198	330	462	594	726	858
ST4_k2	378	630	882	1134	1386	1638
ST4_k3	558	930	1302	1674	2046	2418

Select Suzuki Trotter (order=2)

Hardware constraint exceeded

```
1(g) Experiment hierarchy
|
|-- Choose executable circuit
|   `-- ST2_k2 passes hard constraints
|
|-- Establish fair baselines
|   |-- exact dynamics
|   |-- noiseless ST2_k2 for post-selection
|   `-- ST2_k3 same-depth control for MPF
|
|-- Test mitigation candidates
|   |-- MPF
|   |   |-- works in noiseless algorithmic test
|   |   `-- fails noisy/QPU-readiness test
|   |
|   `-- Post-selection
|       |-- works in noisy simulator
|       |-- works on N=12 QPU
|       `-- cost = shot overhead from acceptance
|
|-- Complete backend observables
|   |-- Z0 for guide 1(g)
|   |-- Z1 and Z0Z1 for PDF 1(h)
|   `-- EstimatorV2 resilience_level=2
|
`-- Check scalability
    |-- N = 12..52
    |— vanilla SamplerV2 and EstimatorV2 res0 measured on QPU
    |— ST2_k2 path stays within 2Q budget
    `— post-selection is resource-scalable, but not yet QPU-measured for every N
```

1) Experiment design

environment \ method	Exact diagonalization	Trotter	MPF	Trotter+Post-sel
Non-noise	①	②	③	—
noise	-	④	⑥	⑤

2) Result comparison matrix

comparison	Measurement variables
① vs ②	Trotter error
② vs ④	Hardware noise error
② vs ③	Improvement with MPF(non-noise)
② vs ④ vs ⑤	Dropped noise by Post-selection
⑥	MPF(noise): variance↑ → error↑(when t is high)

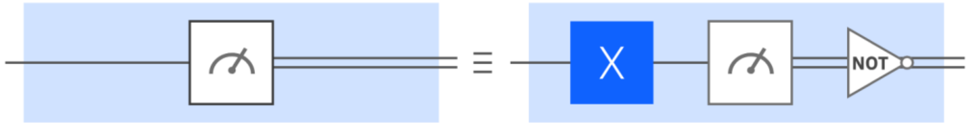
(1) Exact Hamiltonian simulation results
(2) Noiseless Trotter simulation results
(3) Noiseless MPF simulation results
(4) Noisy Trotter simulation results
(5) Noisy Trotter post-selection results
(6) Noisy MPF simulation results

(g) Discuss the trade-off between approximation accuracy and circuit depth. Which construction is best for the same repetition number? Which construction is best for a similar circuit depth?

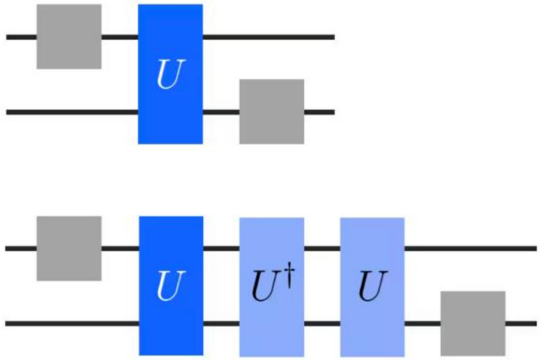
Resilience level

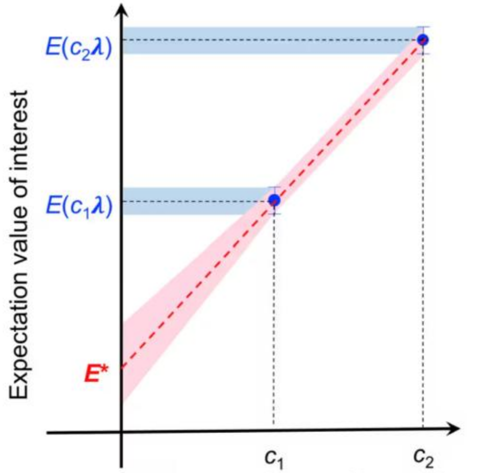
Level	Mitigation Methods
Level 0	None
Level 1 *(Default)*	**TREX**
Level 2	**TREX + ZNE**

TREX



ZNE





In this case, **Level 0 is the best**

(g) Discuss the trade-off between approximation accuracy and circuit depth. Which construction is best for the same repetition number? Which construction is best for a similar circuit depth?

What is MPF

$$f(r) = F + a_1\left(\frac{1}{r^2}\right) + a_2\left(\frac{1}{r^2}\right)^2 + a_3\left(\frac{1}{r^2}\right)^3 + \cdots$$

$$A_{MPF} = \sum_{j=1}^{k} c_j A(r_j),\ \sum c_j \left(\frac{1}{r_j^2}\right)^n = 0\,, n = 1, \ldots, k-1$$

Reps(r)=2, 3

$$f(2) = F + a_1\left(\frac{1}{4}\right) + a_2\left(\frac{1}{16}\right)^2 + \cdots$$

$$f(3) = F + a_1\left(\frac{1}{9}\right) + a_2\left(\frac{1}{81}\right)^2 + \cdots$$

$$S = c_1 f(2) + c_2 f(3) \Rightarrow \text{Canceling } a_1$$

$$c_1 + c_2 = 1$$

$$c_1\left(\frac{1}{4}\right) + c_2\left(\frac{1}{9}\right) = 0 \Rightarrow c_1 = -0.8, c_2 = 1.8$$

Reduce Trotter error

(g) Discuss the trade-off between approximation accuracy and circuit depth. Which construction is best for the same repetition number? Which construction is best for a similar circuit depth?

What is Postselection

"Utilizes known physical symmetries to filter out hardware noise"
($J_x = J_y$ symmetric)

Measure: Obtain bitstring results from the QPU
Verify: Check if the result matches the expected physical state(counting 1's for electrons)
Discard: Completely drop any bitstrings that violate the physical rules.

- more shot needed (~44% bitstrings dropped)

Reduce NISQ hardware error

02 Problem 1

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Results

environment \ method	Exact diagonalization	Trotter	MPF	Trotter+Post-sel
Non-noise	①	②	③	—
noise	-	④	⑥	⑤

1

quantity	<Z0>	<Z1>	<Z0Z1>
Exact diagonalization e^{-iHt}	0.684564	0.991267	0.675831

2

quantity	<Z0>	<Z1>	<Z0Z1>	**vs error**
ST2_k2 (statevector)	0.784583	0.993621	0.778204	**0.100019**

3

quantity	<Z0>	<Z1>	<Z0Z1>	vs exact solution error
MPF23 = (9k3 – 4k2)/5	0.683073	0.991515	0.674588	**0.001491**

4

Shots	<Z0> estimate	Standard Error	vs exact solution	vs non-noise ST2_k2
1024	0.772656	0.0222	0.020514	0.079505
4096	0.722656	0.0117	0.038092	0.061927
8192	0.724854	0.0084	0.040290	0.059729

5

Shots	<Z0> estimate	acceptance	vs non-noise ST2_k2	vs exact solution
1024	0.772404	0.687	0.012179	0.087840
4096	0.791150	0.690	0.006568	0.106586
8192	**0.774869**	0.699	**0.009714**	0.090305

6

Shots	<Z0> estimate	vs exact solution	Non-noise MPF
1024	0.483594	0.200970	0.683073 (error: 0.0015)
4096	0.543359	0.141205	
8192	0.600928	0.083636	

02 Problem 1

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Comparison 1: 1 vs 2 -> Trotter error isolation

1

quantity	<Z0>	<Z1>	<Z0Z1>
Exact diagonalization e^{-iHt}	0.684564	0.991267	0.675831

2

quantity	<Z0>	<Z1>	<Z0Z1>	**vs error**
ST2_k2 (statevector)	0.784583	0.993621	0.778204	**0.100019**

Exact solution 0.684564 vs noiseless Trotter 0.784583
➔ Trotter error=0.100019.
Since there is no noise, all errors are from Trotter

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Comparison 2: 2 vs 4 -> Noise Error check

2

quantity	<Z0>	<Z1>	<Z0Z1>	**vs error**
ST2_k2 (statevector)	0.784583	0.993621	0.778204	**0.100019**

4

Shots	<Z0> estimate	Standard Error	vs exact solution	vs non-noise ST2_k2
1024	0.772656	0.0222	0.020514	0.079505
4096	0.722656	0.0117	0.038092	0.061927
8192	0.724854	0.0084	0.040290	0.059729

Noiseless Trotter 0.784583 vs noise Trotter 0.724854(8192)
➔ Noise error=0.059729
Comparison between in the same circuit with only a noise on/off allows isolation of **pure hardware noise** contribution

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Comparison 3: 2 vs 3 -> MPF improvement

2

quantity	<Z0>	<Z1>	<Z0Z1>	vs error
ST2_k2 (statevector)	0.784583	0.993621	0.778204	**0.100019**

3

quantity	<Z0>	<Z1>	<Z0Z1>	vs exact solution error
MPF23 = (9k3 – 4k2)/5	0.683073	0.991515	0.674588	**0.001491**

Noiseless Trotter error 0.100019
➔ Noiseless MPF error 0.001491 (reduction of Trotter error 98.5%)
MPF extrapolation cancels most of Trotter error(without noise)

02 Problem 1

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Comparison 5: 2 vs 4 vs 5 -> noise reduced by Post-selection

2

quantity	<Z0>	<Z1>	<Z0Z1>	**vs error**
ST2_k2 (statevector)	0.784583	0.993621	0.778204	**0.100019**

4

Shots	<Z0> estimate	Standard Error	vs exact solution	vs non-noise ST2_k2
1024	0.772656	0.0222	0.020514	0.079505
4096	0.722656	0.0117	0.038092	0.061927
8192	0.724854	0.0084	0.040290	0.059729

5

Shots	<Z0> estimate	acceptance	vs non-noise ST2_k2	vs exact solution
1024	0.772404	0.687	0.012179	0.087840
4096	0.791150	0.690	0.006568	0.106586
8192	**0.774869**	0.699	**0.009714**	0.090305

Reduced noise=0.774869-0.724854=0.050015(84% of all noise)
Remaining noise=0.784583-0.774869=0.009714(16% of all noise)

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Explanation -> why does MPF degrade severely with noise

MPF is LC of 1.8 * k3 + (-0.8) * k2. Variance is multiplied by 3.88 times and standard error is amplified 1.97 times. Also, a deeper k3 circuit gets more noise.

These two causes, along with trotter error increasing as time increases causes the severe degradation in MPF reliability. Therefore, in a noisy setting, **Post-selection is more stable than MPF**.

02 Problem 1

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Comparison All

Experiment	<Z0>	vs exact solution	description
Exact solution	0.684564	-	Ideal
Noiseless	0.784583	0.100019	Only Trotter error
Noiseless MPF	0.683073	0.001491	Mose Trotter error canceled
Noise Trotter	0.724854	0.040290	Noise error +0.0597(vs noiseless)
Noise + postselection	0.774869	0.090305	Canceled 84% of noise
Noise MPF	0.600928	0.083636	Failed by variance exploration

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Exact vs noiseless ST2_k2

quantity	value	role
Exact	0.684564	Guide 1(g) exact baseline
Noiseless ST2_k2	0.784583	Trotter-only baseline; +0.100019 bias vs exact
Exact	0.991267	PDF 1(h) observable
Noiseless ST2_k2	0.993621	PF error small: 0.002354
Exact	0.675831	PDF 1(h) observable
Noiseless ST2_k2	0.778204	PF bias: +0.102373
Number-sector leakage	1.35e-14	Weight-1 symmetry preserved; validates post-selection rule

hard-constraint

check	observed	limit	result
depth	63	<100	PASS
2q gates	90	<400	PASS
2q depth	15	<30	PASS
2q survival anchor	(1-0.003)^90 = 0.763	upper-bound diagnostic	reported

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

method	what_it_removes	fair_reference	why
Post-selection	Out-of-sector hardware/noise events	Same ST2_k2 circuit, noiseless Trotter value	It is a hardware-noise filter, not a formula-order correction
MPF23	Trotter bias by extrapolating k=2 and k=3	Unmitigated ST2_k3 raw control	It uses ST2_k2/ST2_k3 and has a higher effective depth/shot cost
resilience_level sweep	Backend mitigation/ZNE-style noise component	Same submitted ST2_k2 circuit, noiseless Trotter value	It modifies execution/mitigation, not the target product formula
Exact dynamics	None; physical target	Exact single-excitation reduction	End-to-end accuracy diagnostic, but can hide Trotter/noise cancellation

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

mitigation	reference	mean_abs_err_vs_reference	mean_abs_err_vs_exact	mean_rel_improve	mean_acceptance	min_accepted_shots	qpu_ready
postselection	noiseless ST2_k2	0.009487	0.094911	0.859377	0.691895	703	True
MPF	exact dynamics / ST2_k3 fair control	0.141937	0.141937	-1.39469	1	2048	False

-> Select: Post selection

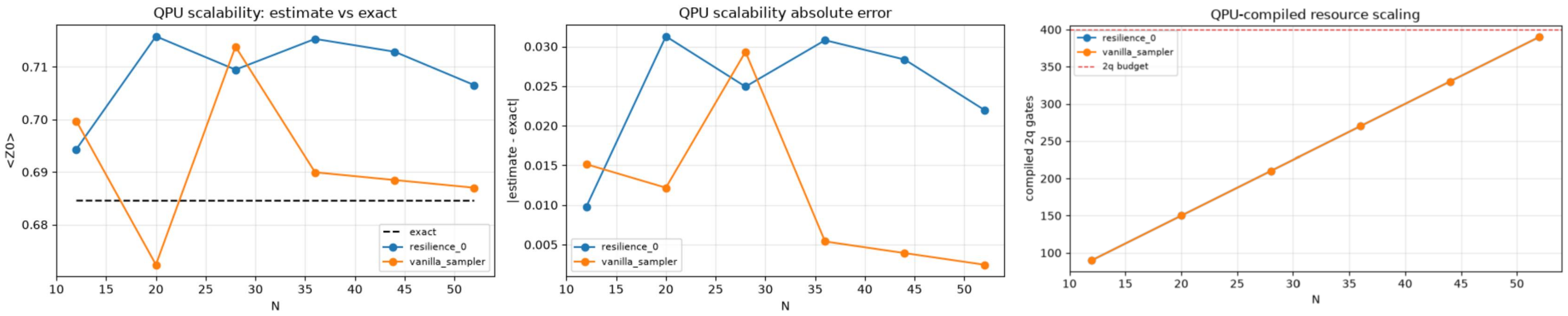
02 Problem 1

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Guide g,h resilience_2, Z0.Z1.Z0Z1 QPU vs exact+absolute error

method	primitive	points	min_N	max_N	mean_abs_err	max_abs_err	max_two_q	budget
resilience_0	EstimatorV2	6	12	52	0.024518	0.031276	390	PASS
vanilla_sampler	SamplerV2	6	12	52	0.011407	0.029323	390	PASS

Problem 1 QPU scalability on eu-de least-busy: vanilla vs resilience-0



(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

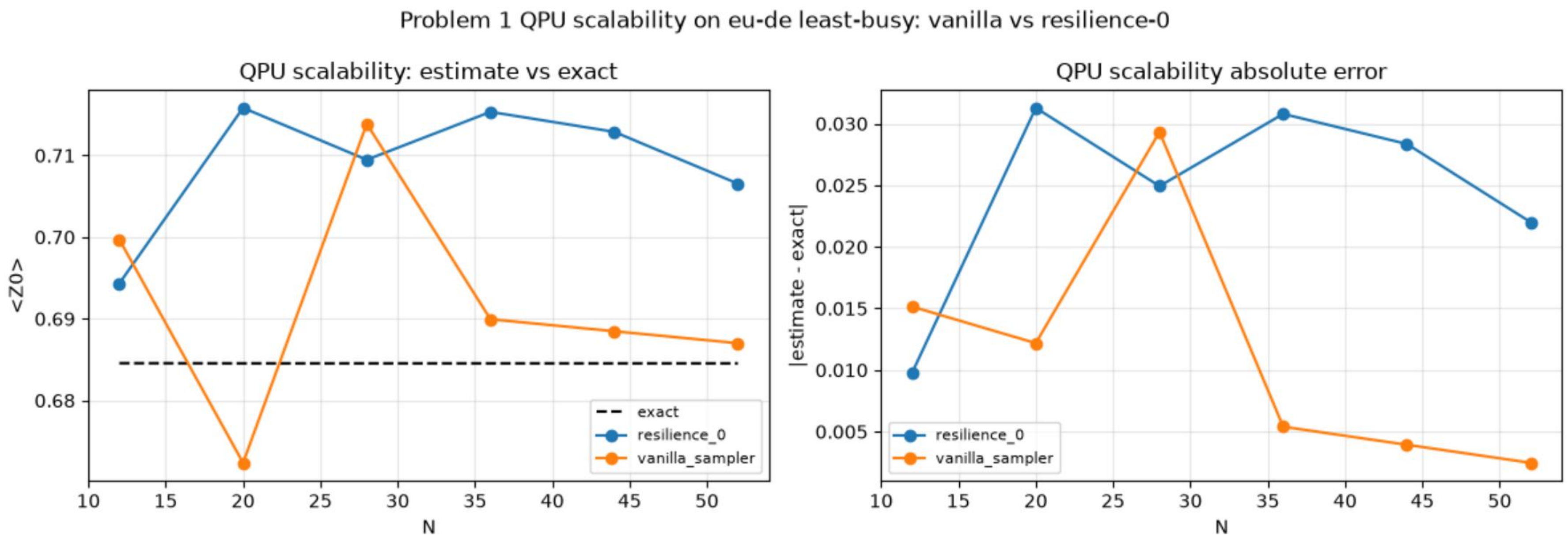
r	N	exact_z0	depth	two_q	within_budget
1	12	0.684564	31	90	True
2	20	0.684544	31	150	True
3	28	0.684544	31	210	True
4	36	0.684544	31	270	True
5	44	0.684544	31	330	True
6	52	0.684544	31	390	True

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

r	N	raw_z0	postselected_z0	acceptance	accepted_shots	raw_err_exact	ps_err_exact	shot_overhead
1	12	0.6997	0.7793	0.666	2728	0.015136	0.094736	1.5
2	20	0.6724	0.7761	0.519	2126	0.012144	0.091556	1.93
3	28	0.7139	0.7975	0.335	1373	0.029356	0.112956	2.98
4	36	0.6899	0.8138	0.205	838	0.005356	0.129256	4.89
5	44	0.6885	0.7821	0.139	569	0.003956	0.097556	7.2
6	52	0.687	0.7833	0.079	323	0.002456	0.098756	12.68

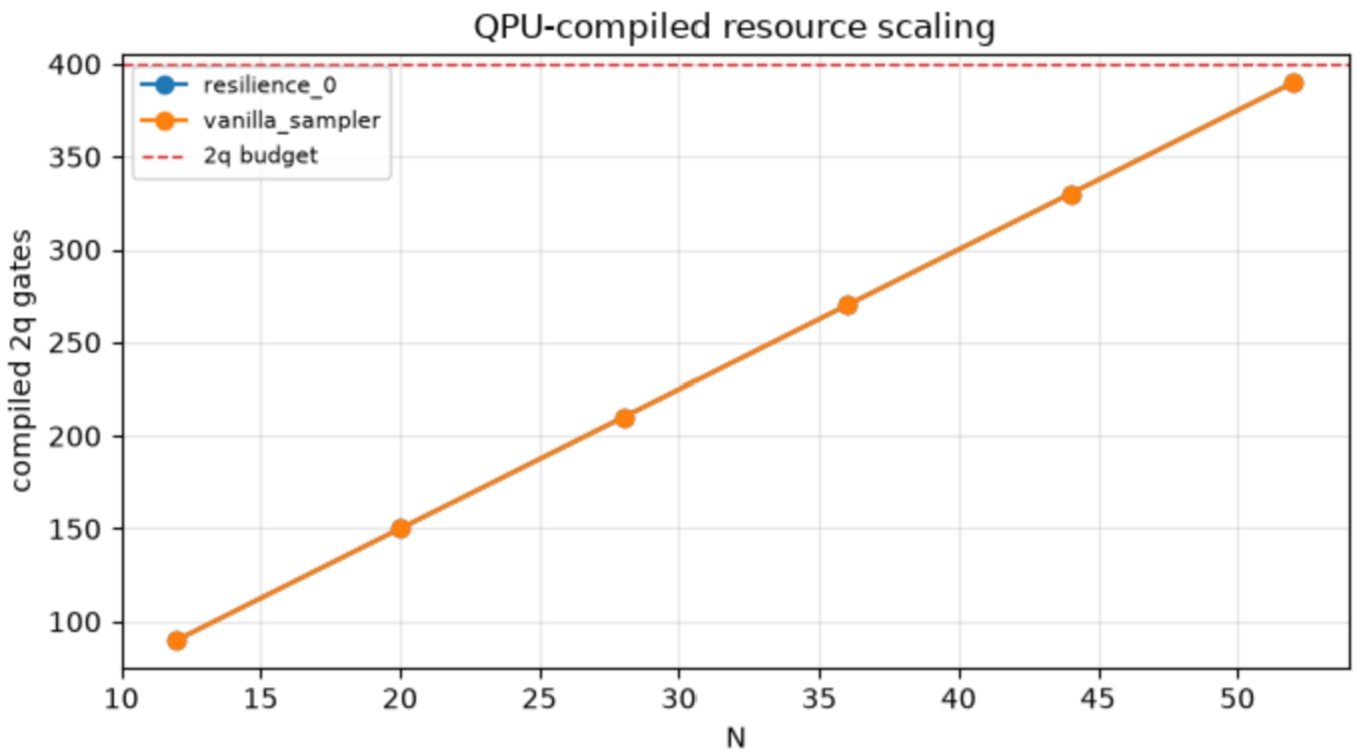
(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Guide g. Z0 scale-up N=12 -> 52 vs exact



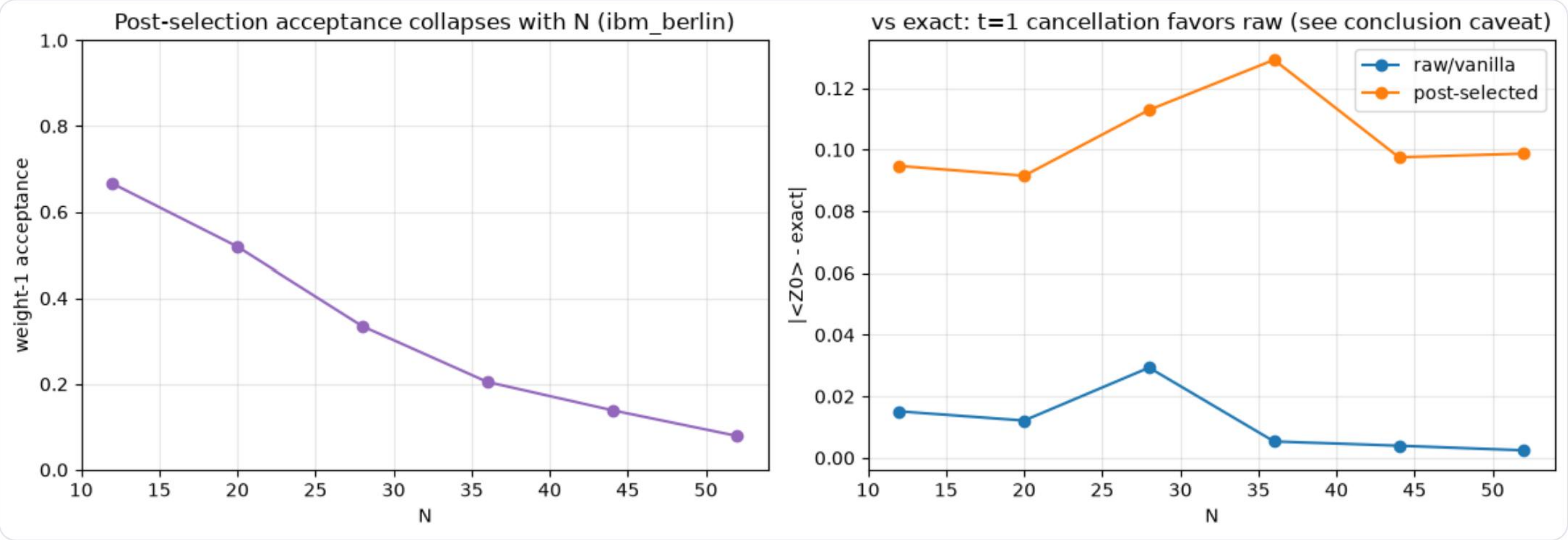
(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

Guide g. 2q vs N

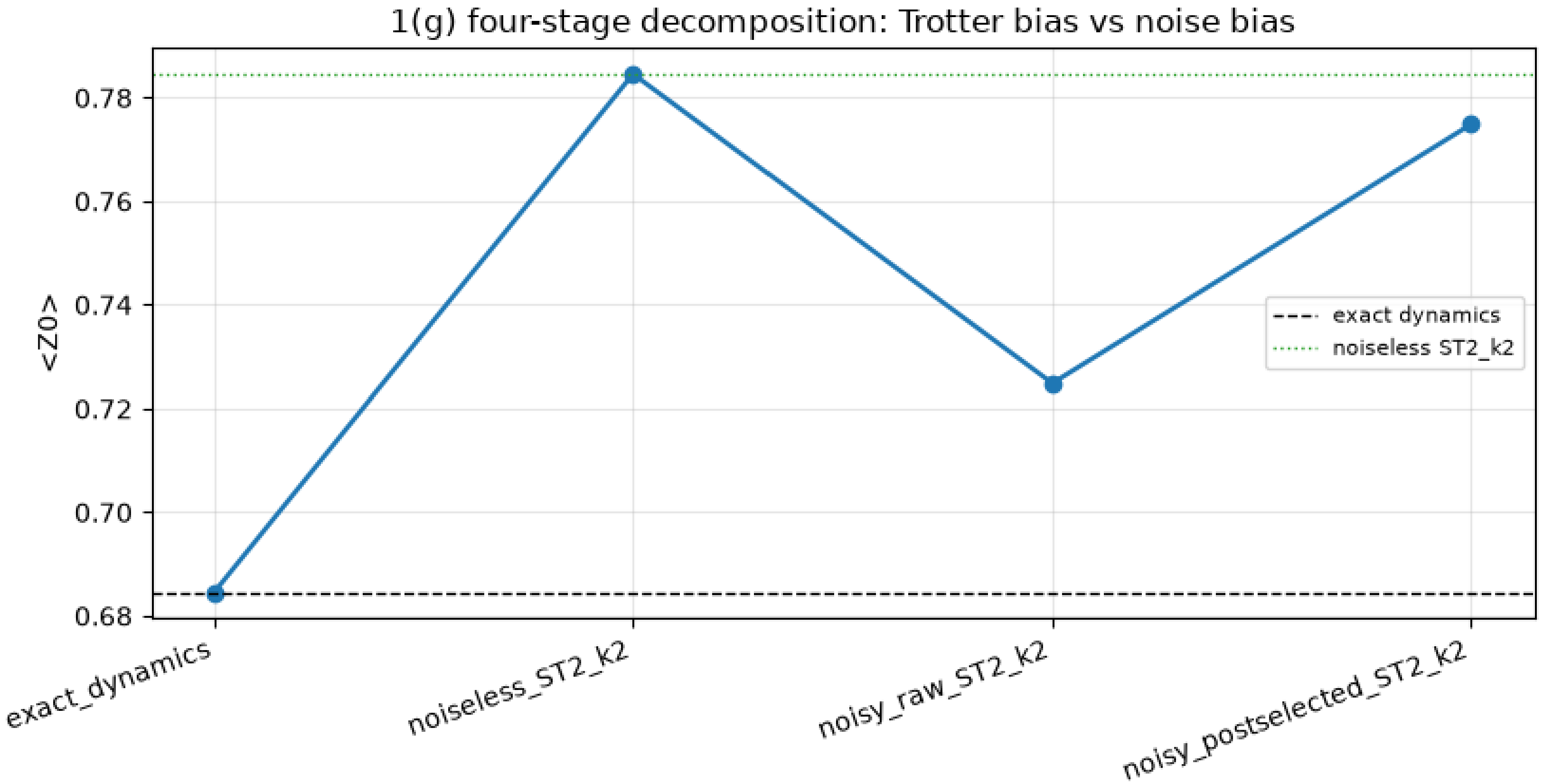


02 Problem 1

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.



(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting 'resilience_level=2' and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.



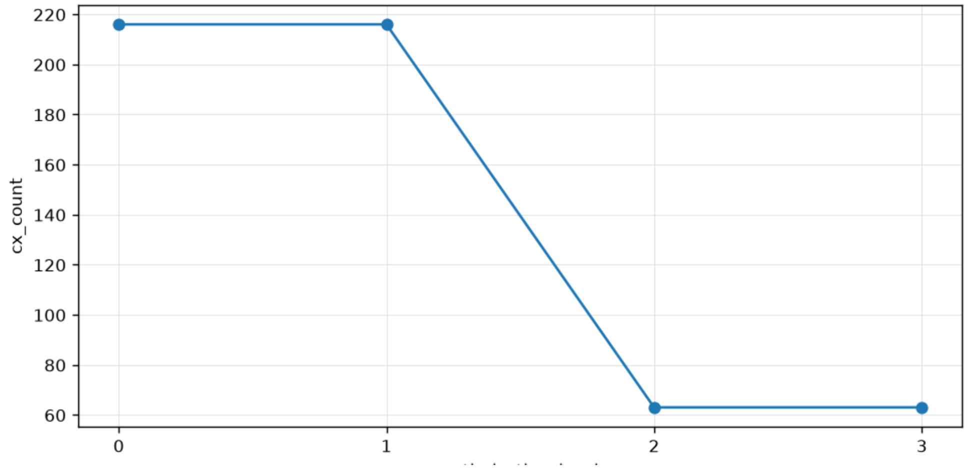
04 Bonus Problem 3: Simple Circuit Optimization

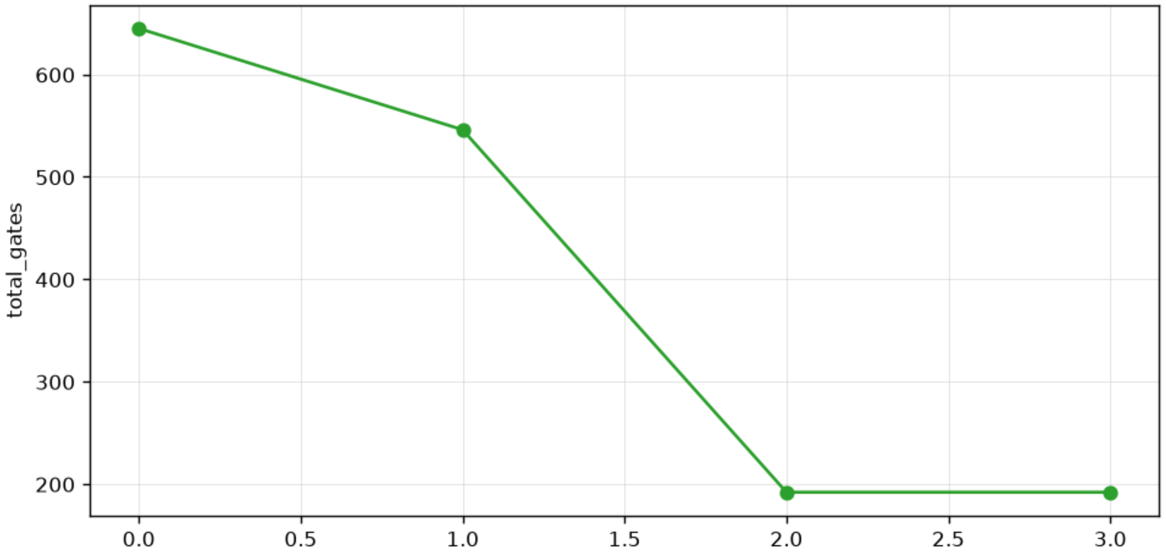
- t=0.5
- N=6
- 2nd Order Suzuki; k=3

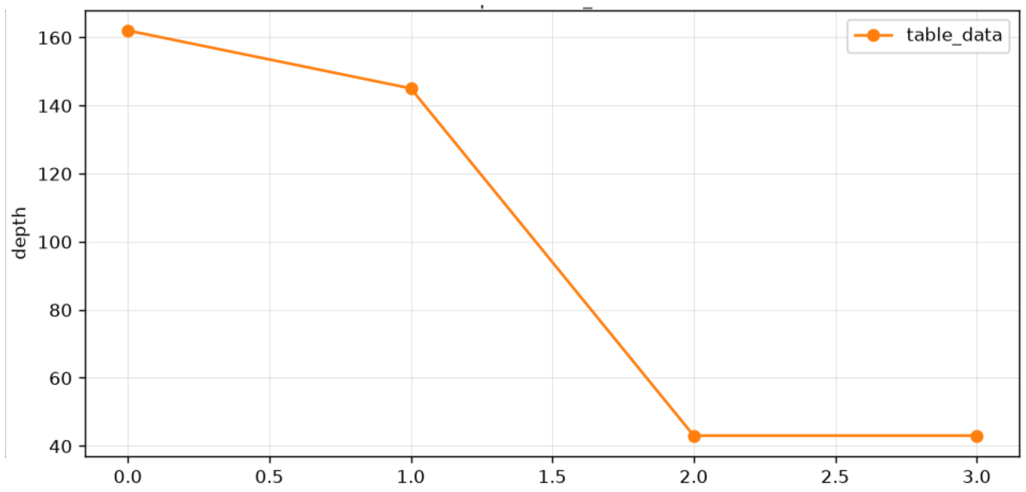
(a) Transpile the circuit for one IBM Quantum backend for fake backend. Compare optimization levels 0,1,2,3.

(b) Report only the essential resource counts: circuit depth, total gate count, and two-qubit gate count.

opt_level	depth	gate count	two-qubit gates
0	162	645	216
1	145	546	216
2	43	192	63
3	43	192	63







- t=0.5
- N=6
- 2^{nd} Order Suzuki; k=1

(c) If available in your environment, try Rustqie-based synthesis for 'PauliEvolutionGate' and compare it with the standard transpilation result.

Further Optimization method: Rustiq?

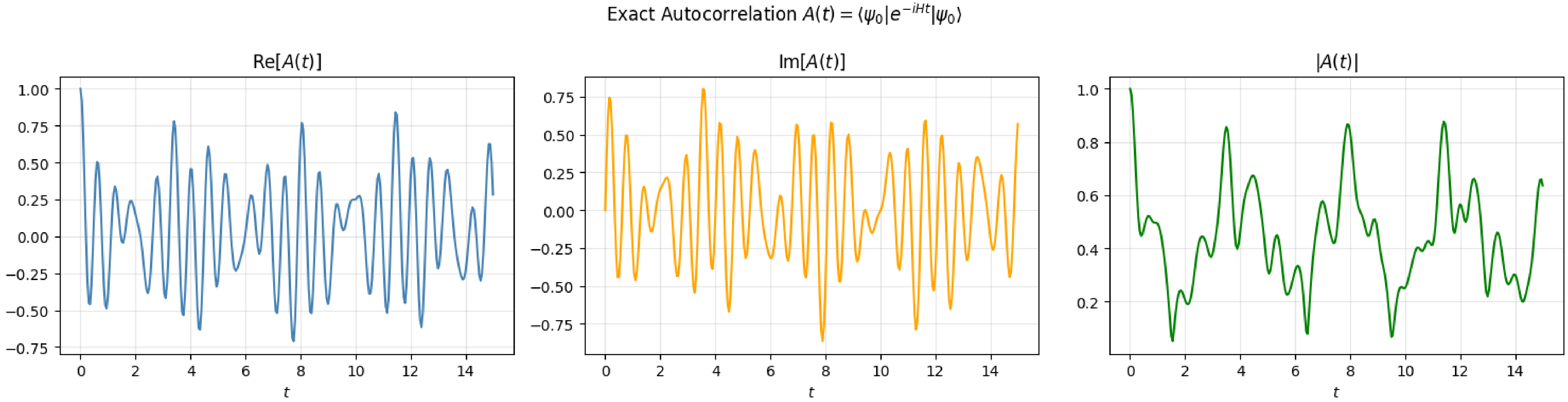
Ordering	Metric	Opt=3	Rustiq (at opt=3)	Opt=0
sequential	depth	67	101	158
	CX	33	78	72
	gates	105	137	215
Even odd	depth	41	157	54
	CX	66	142	72
	gates	150	250	215
Even odd grouped	depth	19	91	54
	CX	27	79	72
	gates	87	149	215

Rustiq Increases Depth & CX for Optimization level 3, but has a moderate optimization effect for Optimization level 0.
➔ Not as good as Opt 3!

Why? ➔ circuit is already locally optimized! Since Hamiltonian is simple, optimization level 3 already creates optimal circuit. Rustiq global re-synthesis instead disrupts structure and increases the entangling-gate cost.

(a) Compute the exact autocorrelation signal $A_{exact}(t)$ on the time grid using classical simulation. Plot its real, imaginary parts and absolute value

Autocorrelation signal: $A(t) = \langle\psi_0|e^{-iHt}|\psi_0\rangle = \Sigma_n|\langle E_n|\psi_0\rangle|^2 e^{-iE_nt}$



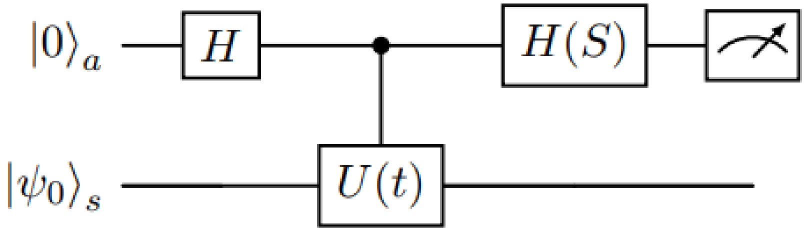
03 Problem 2: Spectral Information from Autocorrelation Signals

- Grouped Ordering
- Synthesis Option: 2nd Order Suzuki; k=3

(b) Estimate the Trotterized autocorrelation signal $A_{PF}(t)$ using at least one product-formula setting from Problem 1 by constructing Hadamard circuits:

(c) Compare $A_{PF}(t)$ with $A_{exact}(t)$. Plot the autocorrelation error $\epsilon_A(\mathrm{t}) = |\mathrm{A}_{\mathrm{exact}}(\mathrm{t}) - \mathrm{A}_{\mathrm{PF}}(\mathrm{t})|$.

$$\frac{1}{\sqrt{2}}(|0\rangle \otimes |\psi_0\rangle + |1\rangle \otimes e^{-iHt}|\psi_0\rangle)$$



$$\rightarrow \frac{1}{2}(|0\rangle + |1\rangle) \otimes |\psi_0\rangle + (|0\rangle - |1\rangle) \otimes e^{-iHt}|\psi_0\rangle)$$

$$\rightarrow |0\rangle \otimes \frac{1}{2}(|\psi_0\rangle + e^{-iHt}|\psi_0\rangle) + |1\rangle \otimes \frac{1}{2}(|\psi_0\rangle - e^{-iHt}|\psi_0\rangle)$$

$$\rightarrow P(0) = \left\|\frac{1}{2}(|\psi_0\rangle + e^{-iHt}|\psi_0\rangle)\right\|^2 = \frac{1}{4}\big(1 + 1 + A(t) + A^*(t)\big) = \frac{1}{2}(1 + Re(A(t)))$$

$$P(1) = \frac{1}{2}(1 - Re(A(t)))$$

$$P(0) - P(1) = Re\big(A(t)\big)$$

I를 취하면

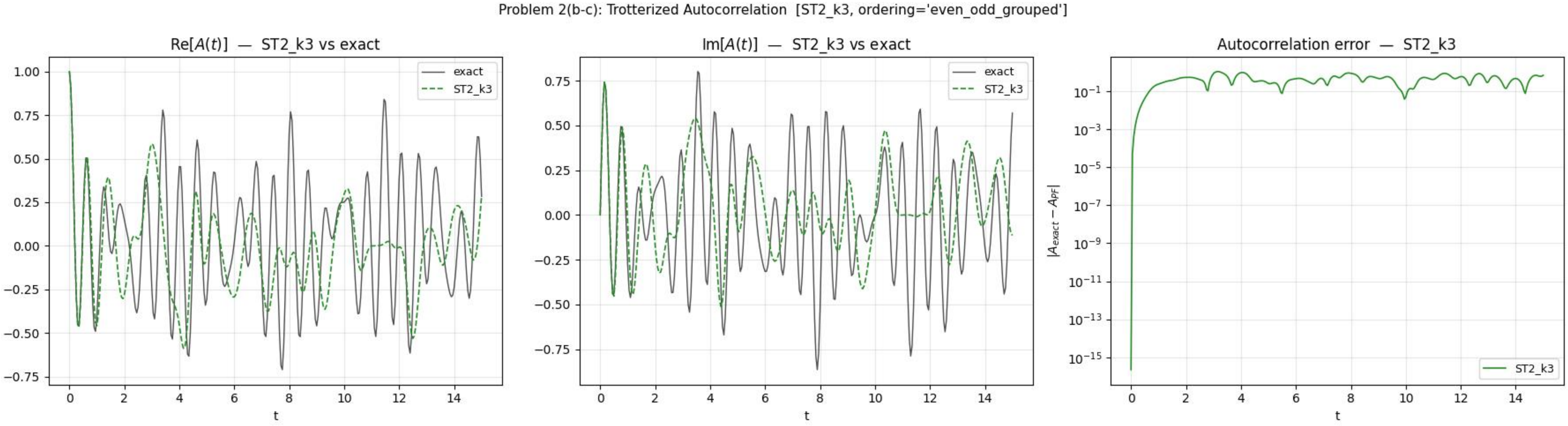
$$P(0) - P(1) = Re(iz) = -Im(A(t))$$

03 Problem 2: Spectral Information from Autocorrelation Signals

- Grouped Ordering
- Synthesis Option: 2nd Order Suzuki; k=3

(b) Estimate the Trotterized autocorrelation signal $A_{PF}(t)$ using at least one product-formula setting from Problem 1 by constructing Hadamard circuits:

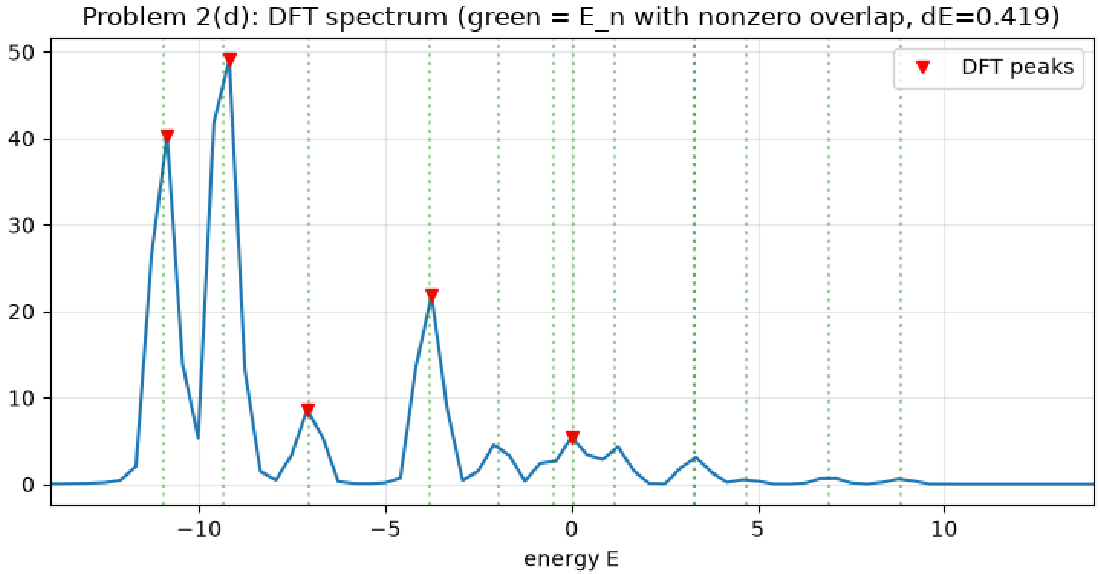
(c) Compare $A_{PF}(t)$ with $A_{exact}(t)$. Plot the autocorrelation error $\epsilon_A(t) = |A_{exact}(t) - A_{PF}(t)|$.

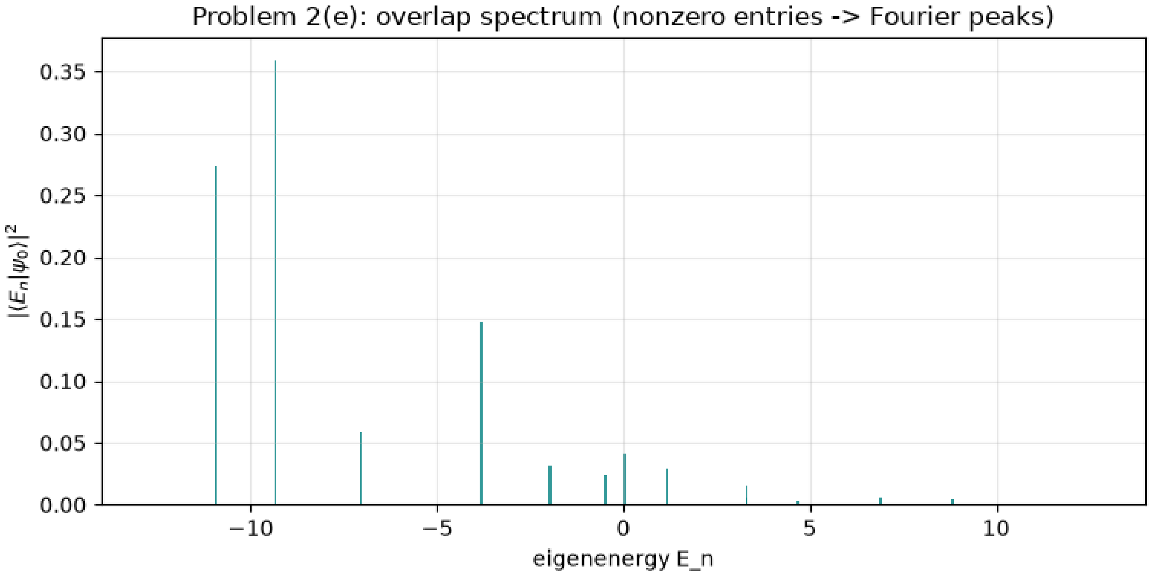


Autocorrelation signal: $A(t) = \langle \psi_0 | e^{-iHt} | \psi_0 \rangle = \Sigma_n |\langle E_n | \psi_0 \rangle|^2 e^{-iE_n t} = \Sigma_n |c_n|^2 e^{-iE_n t}$

(d) Apply a discrete Fourier transform to the exact autocorrelation signal. Plot the resulting spectrum and identify the dominant peaks.

(e) Explain what spectral information is contained in $A(t)$. In particular, discuss why the peak locations are related to eigenenergies that have nonzero overlap with the initial state $|\psi_0\rangle$.



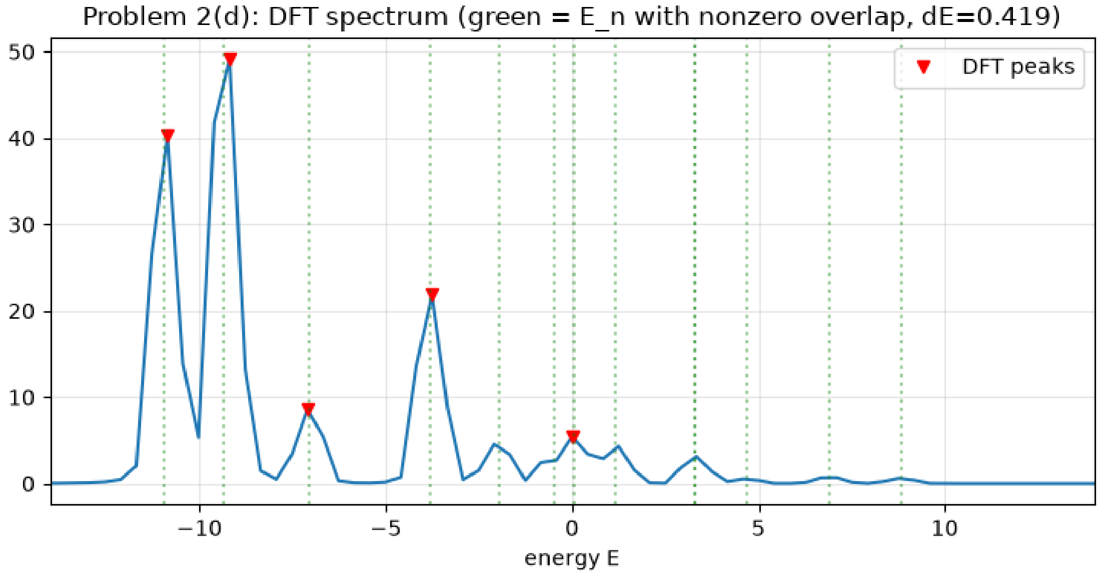


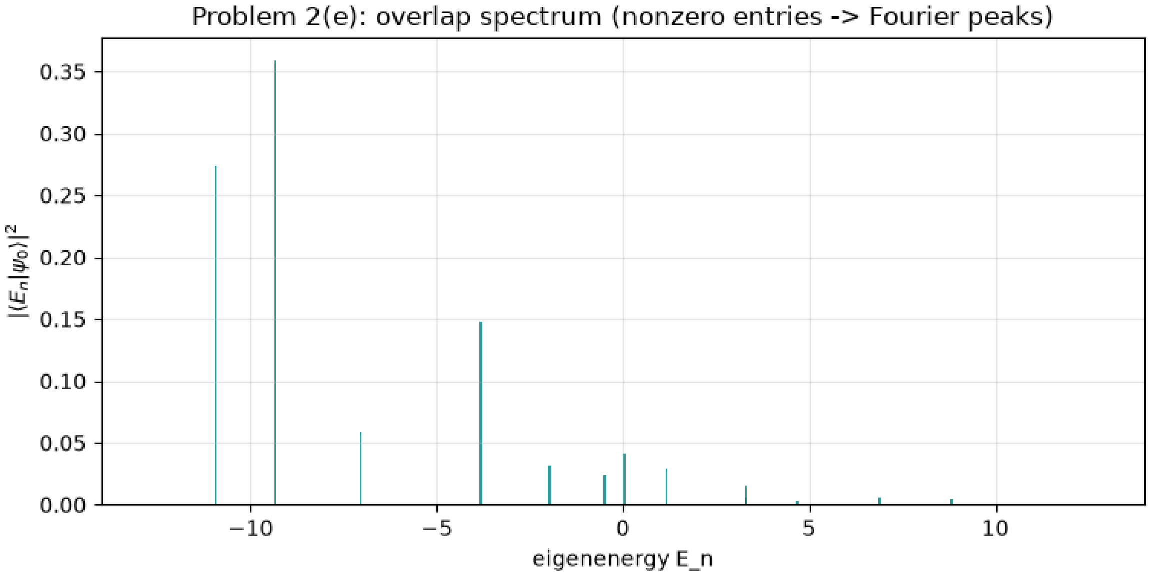
Peak energy	Nearest eigenvalue	Overlap weight
-10.855	-10.937	0.274
-9.185	-9.341	0.359
-7.097	-7.031	0.058
-3.757	-3.819	0.148
0	0.033	0.041

Autocorrelation signal: $A(t) = \langle \psi_0 | e^{-iHt} | \psi_0 \rangle = \Sigma_n |\langle E_n | \psi_0 \rangle|^2 e^{-iE_n t} = \Sigma_n |c_n|^2 e^{-iE_n t}$

(d) Apply a discrete Fourier transform to the exact autocorrelation signal. Plot the resulting spectrum and identify the dominant peaks.

(e) Explain what spectral information is contained in $A(t)$. In particular, discuss why the peak locations are related to eigenenergies that have nonzero overlap with the initial state $|\psi_0\rangle$.





Since only eigenenergies with nonzero overlap with $|\psi_0\rangle$ have a nonzero c_n, if we do DFT, related eigenenergies create peaks with size corresponding to value of c_n.

➔if c_n is too small, eigenenergy may not be detected.

(a) Construct Hamiltonian using ‘SparsePauliOp’. Report total number of Pauli terms and identify which terms come from the periodic boundary condition.

(b) Construct product-formula circuits for each target time t in the time grid. For each t, directly approximate $U(t) = e^{-iHt}$ using ‘PauliEvolutionGate’ with given synthesis options for $k = 1, 2, 3$

(c) Decompose or transpile each circuit into the basis gates $\{cx, u3, u1\}$. For each synthesis option, report circuit depth, total gate count, number of cx gates, and number of one-qubit gates.

(d) Use exact classical simulation for $N = 6$ to obtain the reference state $|\psi_{exact}(t)\rangle = e^{-iHt}|\psi_0\rangle$. Compare it with the product-formula state $|\psi_{PF}(t)\rangle$ using the state infidelity $\epsilon_{state}(t) = 1 - |\langle\psi_{exact}(t)|\psi_{PF}(t)\rangle|^2$.

(e) Compute local-observable errors for Z_1, Z_0Z_1. For an observable O, use $\epsilon_O(t) = |\langle O\rangle_{exact}(t) - \langle O\rangle_{PF}(t)|$.

(f) Plot time versus error for the state infidelity, Z_1 error, and Z_0Z_1 error. Compare Lie-Trotter, 2nd order Suzuki-Trotter, and 4th order Suzuki-Trotter formulas.

(g) Discuss the trade-off between approximation accuracy and circuit depth. Which construction is best for the same repetition number? Which construction is best for a similar circuit depth?

(h) Run a small subset of the circuits on an available IBM Quantum backend. Choose a few representative time points from the time grid and estimate $\langle Z_1\rangle$ and $\langle Z_0Z_1\rangle$ using one product-formula construction. Use Qiskit Runtime error mitigation, for example by setting ‘resilience_level=2’ and compare the back end results with the ideal exact-simulation values. Discuss the main sources of discrepancy.

(a) Compute the exact autocorrelation signal $A_{exact}(t)$ on the time grid using classical simulation. Plot its real, imaginary parts and absolute value

(b) Estimate the Trotterized autocorrelation signal $A_{PF}(t)$ using at least one product-formula setting from Problem 1 by constructing Hadamard circuits:

(c) Compare $A_{PF}(t)$ with $A_{exact}(t)$. Plot the autocorrelation error $\epsilon_A(\mathrm{t}) = |\mathrm{A_{exact}}(\mathrm{t}) - \mathrm{A_{PF}}(\mathrm{t})|$.

(d) Apply a discrete Fourier transform to the exact autocorrelation signal. Plot the resulting spectrum and identify the dominant peaks.

(e) Explain what spectral information is contained in $A(t)$. In particular, discuss why the peak locations are related to eigenenergies that have nonzero overlap with the initial state $|\psi_0\rangle$.

(a) Transpile the circuit for one IBM Quantum backend for fake backend. Compare optimization levels 0,1,2,3.

(b) Report only the essential resource counts: circuit depth, total gate count, and two-qubit gate count.

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